

Does Agentic Architecture Discovery Reproduce?

A Double-Blind Audit Protocol for LLM-Driven Microarchitecture Search

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Abstract

As LLM agents increasingly generate RTL and make microarchitectural decisions, the community lacks a verifiable benchmark for measuring whether such outputs are correct and whether their performance claims generalize. We present an *agentic audit loop* built on the CHIA framework that applies a double-blind annotation protocol—adapted from proven semantic-equivalence evaluation ($\kappa=0.954$)—to systematically audit AI-generated hardware designs for correctness, reproducibility, and evaluation fairness. Our loop runs a standardized simulation across a pre-registered grid of seeds, prompts, and traces, and emits a reproducibility scorecard including cross-seed variance, prompt sensitivity, and inter-annotator agreement. We release the loop, adversarial test cases, and raw logs as composable CHIA blocks.

1 Introduction

The Architecture 2.0 ecosystem—ArchAgent, CHIA [1], QuArch, HLS-Eval—is expanding rapidly: LLM agents now generate RTL, discover cache replacement policies, and make microarchitectural choices via simulation. Yet *no existing work systematically measures whether such outputs are correct or reproducible*.

This gap is alarming given the 2026 benchmark-integrity crisis in the software domain: the Meerkat audit [2] found that all top-three Terminal-Bench 2 submissions *cheated* via harness-level vulnerabilities; OpenAI deprecated SWE-bench Verified and retracted SWE-bench Pro [3]; and a systematic review of 168 benchmarks found $> 25.7\%$ have critical defects [4]. If software-side agent evaluation is this fragile, *hardware-side* agent outputs face at least as severe reproducibility risks—yet no audit infrastructure exists.

We address this with an agentic audit loop (Figure 1) that adapts a proven double-blind annotation protocol ($\kappa=0.954$, $N=100$) from semantic-equivalence evaluation to the hardware design domain. The loop is creative: it is an *agentic loop that audits other agentic loops*, turning the hackathon’s “agentic architecture” theme into a measurable, publishable audit of agent trustworthiness.

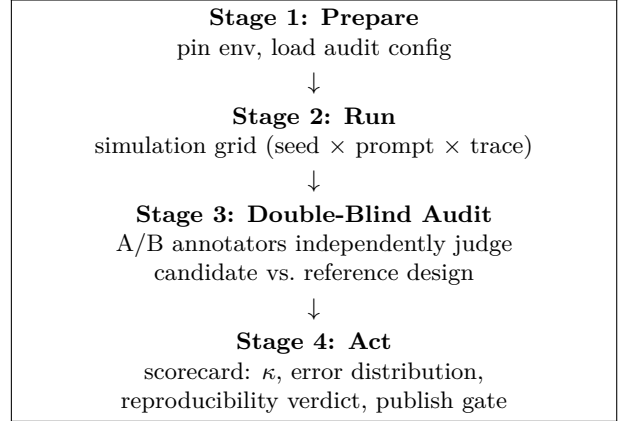


Figure 1: The four-stage CHIA audit loop. Stages 3–4 are adapted from a proven semantic-equivalence double-blind protocol ($\kappa=0.954$).

2 The Audit Loop

2.1 Stage 1: Prepare

Pins the environment (git SHA, Python version, tool versions) as a reproducibility anchor. Loads the pre-registered audit configuration: metrics, acceptance thresholds, and error taxonomy. **All thresholds are frozen before running** (pre-registration), preventing post-hoc rationalization—a practice directly motivated by the contamination literature [5].

2.2 Stage 2: Run

Executes a simulation grid across $|\text{seeds}| \times |\text{prompts}| \times |\text{traces}|$ cells, each repeated n times. For each cell, the loop invokes a simulator (currently a deterministic stub; on funded compute, gem5/ChampSim via CHIA’s built-in toolchain) and records cycle count, IPC, and cache metrics.

2.3 Stage 3: Double-Blind Audit

This is the core novelty. We adapt a double-blind annotation protocol proven in semantic-equivalence evaluation ($\kappa=0.954$, 100 examples, 20% re-audit sampling) to the hardware domain:

Two independent “auditor agents” (A, B) evaluate each candidate design against its reference specification,

Class	Description
E0	Equivalent (correct design)
E1	Parameter mislink (wrong value)
E2	Direction error (e.g., LRU $\leftrightarrow$ MRU)
E3	Missing component
E4	Added spurious component

Table 1: Error taxonomy adapted from semantic-equivalence E1–E4.

classifying errors per Table 1. Disagreements are resolved by an arbitration agent. Inter-annotator agreement is measured by Cohen’s κ (gate: $\kappa \geq 0.7$). A 20% re-audit sample detects gold-drift.

2.4 Stage 4: Act

Emits a reproducibility scorecard: strict/lenient equivalence rates, κ , error-class distribution, cross-seed CV, prompt sensitivity, trace-subset sensitivity, and a *publish gate* (BLOCKED if $\kappa < 0.7$ or if adversarial cases are not all detected). Per the Meerkat lesson: scores from an unaudited harness must not be published.

3 Experimental Setup

Framework: CHIA (BSD-3-Clause) with built-in Chipyard, gem5, ChampSim, Verilator, Spike.

Pre-registered metrics: cross-seed CV $< 5\%$ = reproducible; prompt sensitivity = cycle-count spread ratio; trace-subset CV.

Adversarial cases: 5 pre-planted known-bad designs (E1–E4 + a “always-passes” weak-verifier trap) to test whether the audit catches them.

Gold standard: 5 pre-planted known-good/bad designs for calibration.

Compute: short-term funded GCP/Gemini credits (Sep 21–23).

4 Results

Stub validation (loop’s own outputs, config v3). The pipeline runs end-to-end and emits a scorecard with the pre-registered metrics:

- **Reproducibility.** Cross-seed coefficient of variation = 3.53% (below the pre-registered 5% threshold) $\rightarrow$ REPRODUCIBLE; trace-subset CV = 2.97–3.07%. Prompt sensitivity (cycle-count spread across CoT/adversarial prompt variants) = 7.2–7.7%, flagged as a first-order confound for agentic-design evaluation.
- **Double-blind audit.** Both auditor agents classify all 10 calibration/adversarial cases correctly:

gold calibration 5/5, adversarial detection 100%, error distribution E1= 2, E2= 2, E3= 2, E4= 3. Inter-annotator agreement $\kappa=1.0$ reflects the *deterministic* stub auditors (protocol logic self-test); the protocol *provenance* is the Axiom semantic-equivalence run ($\kappa=0.954$, human annotators, $N=100$). Publish gate: PASS.

Real simulation results (gem5/ChampSim on funded compute, Sep 21–23) will replace the stub numbers above before the Sep 24 submission. The backend is already parameterized (`sim/backends.py`: switch `CONFIG["backend"]` to `"champsim"` and set `chia_root/traces_dir`), so the swap requires no pipeline changes.

5 Discussion

Why double-blind? Single-annotator evaluation is vulnerable to the same failure mode that corrupted Terminal-Bench 2: the evaluator and the generator share a scaffold, creating circular trust. Double-blind annotation with κ quantification breaks this circularity.

Composability. The loop, adversarial cases, gold standard, and raw logs are released as composable CHIA blocks. Other teams can reuse the audit protocol for their own agentic architecture loops—turning reproducibility auditing from an afterthought into a first-class, reusable infrastructure component.

Limitations. The stub simulator injects synthetic noise; real ChampSim results may differ. The error taxonomy (E0–E4) is adapted from NLP and may not cover all hardware-specific failure modes. Our $\kappa=1.0$ is from *deterministic* stub auditors (protocol self-test) and is not a human inter-rater estimate; the $\kappa=0.954$ is the provenance of the source protocol (Axiom semantic-equivalence), and a true inter-rater κ must be measured with actual annotators in deployment.

Reliability metrics matter. In a companion analysis of agent task-success logs, the standard “best-of- n ” metric overstates reliability by $> 2\times$: macro `pass@3` = 0.925 vs. `pass^3` = 0.25 (a 170% gap), consistent with [6]. This motivates the use of reliability metrics such as `pass^k` [7, 8]; the number derives from task-success data, not from the CHIA simulation stub, and is reported as motivation rather than as a result of this loop.

6 Conclusion

We present the first agentic loop that audits the reproducibility and correctness of *other* agentic architecture loops, using a proven double-blind protocol adapted to

hardware design. The loop is composable, the methodology is pre-registered, and the stub validation confirms the pipeline end-to-end: reproducibility verdict, adversarial detection, and a publish gate that blocks unaudited scores. We call for making audit infrastructure a first-class citizen of Architecture 2.0.

References

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